

ALGORITHM

Autumn 2026, Quiz 4, March 25

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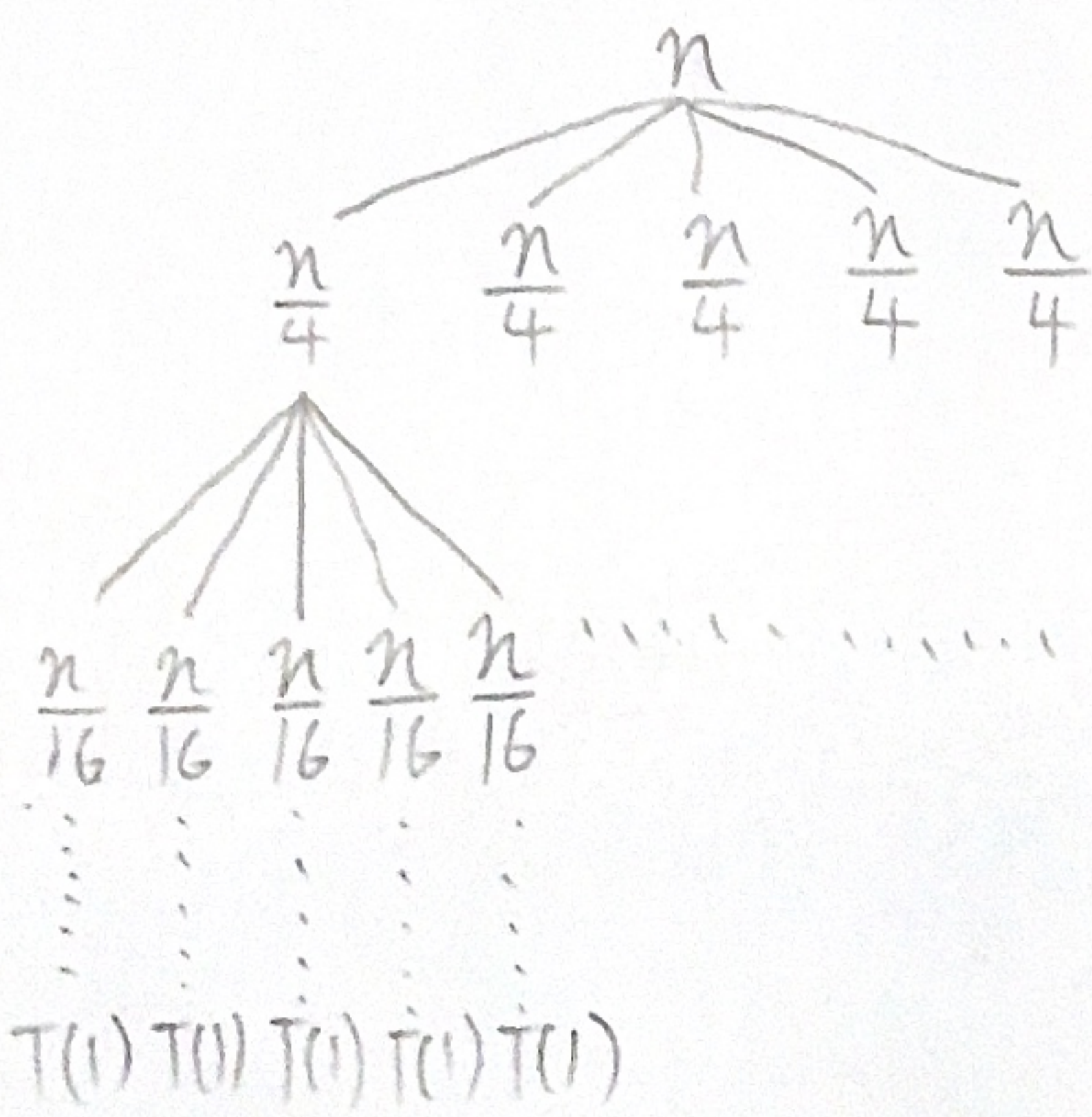
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1. Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 5T(\lfloor n/4 \rfloor) + n$, where $T(1) = 1$, $T(2) = 2$, and $T(3) = 3$. Use the substitution method to verify your answer.

$$T(n) = 5T(\lfloor n/4 \rfloor) + n, T(1)=1, T(2)=2, T(3)=3$$



每層花費	層數
n	0
$\frac{n}{4} \times 5$	1
$\frac{n}{16} \times 5^2$	2
$\vdots$	$\vdots$
$\frac{n}{4^{h-1}} \times 5^{h-1}$	$h-1$

假設遞迴樹共 h 層:						
層數	0	1	2		$h-1$	
子問題大小	n	$\frac{n}{4}$	$\frac{n}{16}$		$\frac{n}{4^{h-1}}$	

$$\frac{n}{4^{h-1}} = 1, n = 4^{h-1}, h = \log_4 n + 1$$

$$\text{第 } k \text{ 層 cost: } \frac{n}{4^k} \times 5^k = \left(\frac{5}{4}\right)^k n$$

整棵樹 total cost:

$$n + \frac{5}{4}n + \frac{25}{16}n + \dots + \left(\frac{5}{4}\right)^{\log_4 n} n$$

$$= \sum_{i=0}^{\log_4 n} \left(\frac{5}{4}\right)^i n = n \times \frac{\left(\frac{5}{4}\right)^{\log_4 n + 1} - 1}{\frac{5}{4} - 1} = 4n \times \left[\left(\frac{5}{4}\right)^{\log_4 n + 1} - 1\right]$$

$$= 4n \left[\frac{5^{\log_4 n + 1}}{4^{\log_4 n + 1}} - 1 \right] = 4n \left[\frac{n^{\log_4 5}}{n} - 1 \right]$$

$$= 4n^{\log_4 5} - 4n, \Rightarrow T(n) = O(n^{\log_4 5})$$

Substitution method verify:

pf that $T(\lfloor \frac{n}{4} \rfloor) \leq cn^{\log_4 5} - 5n$

why? -1

Basic Step:

$$T(1) = 1 \leq c_1 \times 1^{\log_4 5} - 5, c_1 \geq 6$$

$$T(2) = 2 \leq c_1 \times 2^{\log_4 5} - 10, c_1 \geq 6$$

$$T(3) = 3 \leq c_1 \times 3^{\log_4 5} - 15, c_1 \geq 6$$

$$\rightarrow c_1 = 6, n_0 = 1$$

Induction Step:

assume that $T(\lfloor \frac{n}{4} \rfloor) \leq c_2 \left(\frac{n}{4}\right)^{\log_4 5} - 5n$ for $\lfloor \frac{n}{4} \rfloor$

$$\begin{aligned} \text{then, } T(n) &= 5T(\lfloor \frac{n}{4} \rfloor) + n \\ &\leq 5c_2 \left(\frac{n}{4}\right)^{\log_4 5} - 5n + n \\ &\leq 5c_2 \left(\frac{n}{4}\right)^{\log_4 5} - 4n \\ &\leq c_2 n^{\log_4 5} \end{aligned}$$

當 $c_2 > 0$ 時, 恒成立

$$\therefore \exists c = \max(c_1, c_2) = 6 > 0, n_0 = 1, \text{ s.t. } T(n) \leq cn^{\log_4 5} \forall n \geq n_0$$